

1 **Reinforcing kōura resilience using rock piles to**
2 **enhance freshwater crayfish habitat in a**
3 **Aotearoa New Zealand lake**

4 **Stone piles Rotoiti**

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7 2026-07-07

8 Placeholder — abstract to be written.

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Introduction

Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater fauna now threatened with extinction (Sayer *et al.*, 2025). Among the most endangered are freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss, invasive species, and water quality decline (Richman *et al.*, 2015). As keystone species and ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and foodweb dynamics, and their loss can have cascading consequences for freshwater ecosystems (Momot, 1995; Collier, Parkyn & Rabeni, 1997; Parkyn, Collier & Hicks, 2001).

In Aotearoa New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura (*Paranephrops planifrons* White, 1842) is a taonga (culturally treasured) species of significant ecological and cultural importance to Māori, particularly for Te Arawa iwi of the Rotorua lakes region in the North Island of Aotearoa (Kusabs & Quinn, 2009). Kōura support customary fisheries using cultural practices such as tau kōura, yet populations in Lakes Rotoiti and Rotorua have declined by up to 96 % over the past two decades, with consequent loss of these customary practices (Kusabs *et al.*, 2015a; Kusabs *et al.*, 2026). The primary driver of decline is predation by the invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819, hereafter catfish), confirmed in these lakes in 2016 and 2018 (Francis, 2019; Grayling, 2020). Catfish are native to eastern North America where they mostly live in muddy slow-moving water bodies and can survive and spawn in a wide temperature range, tolerate poor water quality, low oxygen concentration and have a broad diet (Scott & Crossman, 1973), making them a highly adaptable/tolerant invader. Catfish were introduced in Aotearoa in 1877 as sport fish (McDowall, 1994) with their distribution continuously spreading over the upper North Island such as the Waikato River and in 1985 they were detected in Lake Taupō (Barnes & Hicks, 2003). In Aotearoa catfish are a major predator on kōura, as 64 % of large catfish (<250 mm fork length) had kōura in their stomach contents (Barnes & Hicks, 2003). In combination with predation by catfish kōura are further compressed by multiple interacting stressors that progressively restrict their available habitat. Kōura are naturally found in complex rocky habitats (Jowett, Parkyn & Richardson, 2008; Kusabs, Quinn & Hamilton, 2015b) or in deeper parts of the lake to avoid predation during the day, migrating to the productive littoral zone at night to forage (Devcich, 1979). However, thermal stratification creates anoxic conditions in deeper waters, displacing kōura into the shallow littoral zone

(Kusabs *et al.*, 2026), precisely where catfish predation pressure is greatest, as catfish rarely swim beyond 17 m depth (Dedual, 2002). This compression is further intensified by dense invasive macrophyte beds obstructing kōura migration in the littoral zone (Kusabs & Quinn, 2009), which also intensify oxygen depletion in deeper waters by die-off events (Vincent, Gibbs & Dryden, 1984; Hamilton, McBride & Uraoka, 2005). The importance of deep refuge is shown by Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake remains oxygenated below the thermocline (Verburg & Albert, 2019). In Lake Rotoiti this deep refuge becomes anoxic and unavailable for kōura (Kusabs *et al.*, 2026). Next to direct predation catfish exposure triggers anti-predator responses in kōura including increased refuge use (Raven *et al.*, (a) in review), reducing time available for foraging and potentially suppressing growth and reproduction (Preisser, Bolnick & Benard, 2005; Sheriff *et al.*, 2020). Climate change is expected to strengthen thermal stratification, expand anoxic zones (Woolway *et al.*, 2021), and increase the range of suitable catfish habitat in Aotearoa, further reducing kōura persistence (Lee *et al.*, 2025).

The compression of kōura into the littoral zone makes the quality of nearshore habitats increasingly important for kōura populations. In stream environments, kōura are positively associated with structurally complex habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Usio & Townsend, 2000; Parkyn & Collier, 2004; Jowett *et al.*, 2008; Parkyn, Meleason & Davies-Colley, 2009). In lake environments benthic substrate characteristics are a stronger predictor of kōura distribution than nutrient enrichment or predator abundance (Kusabs *et al.*, 2015b), and kōura are positively associated with coarse substrates and riparian vegetation (Raven *et al.*, (b) in review). Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish predation on crayfish (Adams, 2007; Zhao *et al.*, 2024), and kōura increase refuge use in the presence of catfish compared to native predators (Raven *et al.*, (a) in review), emphasising the importance of shelter availability for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds of introduced macrophytes and little natural rocky structure. Adding rocky substrate to soft-bottomed lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl, 2008), and increasing structural complexity in streams improves crayfish restocking success (Huolila, Marjomäki & Laukkanen, 1997). Deploying habitat enhancement structures that add structural complexity in the littoral zone could potentially be a practical way to restore kōura populations in Lake Rotoiti.

18 In this study habitat enhancement structures in the shape of stone piles were deployed along
19 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
20 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
21 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
22 kōura occupancy, abundance and biomass would increase at stone pile sites relative to control
23 sites after construction, and that catfish would not change. We further predicted that kōura
24 abundance would keep increasing progressively after construction, at least within the first 18
25 months, and that catfish would instead show seasonal patterns of shoreline use rather than a
26 construction-driven trend (Barnes, 1996; Dedual, 2002). Finally, we predicted that stone pile
27 sites would progressively converge toward kōura densities observed at natural rocky reference
28 sites, and that catfish would not show this pattern.

29 **Materials and Methods**

1 **Study location and construction**

2 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
3 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
4 stratifies for approximately nine months a year. Since 2008, nutrient-rich inflows from Lake
5 Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to
6 improve water quality in Lake Rotoiti (Hamilton *et al.*, 2005; Prentice & Özkundakci, 2024).
7 Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and
8 controlled water level throughout the year. Catfish are present throughout the lake but occur
9 in particularly high concentrations in Te Weta Bay on the North West side of the lake (Francis,
10 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and
11 land owner approval (Figure 1). At each stone pile site, a paired control site was established
12 50 m along the shoreline in the same habitat type. Five naturally occurring rocky shoreline
13 locations were selected as reference sites, each with two monitoring points 50 m apart. In
14 total 20 sites of 150 m² (15 * 10 m) were monitored seasonally over 18 months. Baseline
15 monitoring was conducted in February 2025 prior to stone pile deployment, which occurred
16 the following week. Stone piles were constructed from locally sourced quarry rock (Manawahe
17 Stone and Quarry, ~25 km from the lake) with a size range of 50 - 400 mm. At each of the
18 five stone pile sites, three stone piles were arranged horizontally along the shoreline by hand
19 from the shore. Stone pile design was informed by mesocosm experiments demonstrating kōura

20 preference for enclosed stone structures (Raven *et al.*, (c) in review). Stone piles were positioned
21 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

22 Monitoring

23 Monitoring consisted of physical, chemical, and biological parameters recorded at each site
24 during each monitoring round. Static site characteristics including substrate composition, slope
25 to depth contour, riparian vegetation, overhanging tree cover and wood cover were visually
26 estimated at baseline monitoring only, as these were not expected to change substantially over
27 the monitoring period. Substrate composition was visually estimated using an underwater
28 viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64
29 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter. A substrate index was
30 calculated from these estimates (Jowett & Richardson, 1990; Jowett *et al.*, 2008):

$$\text{Substrate index} = 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} + 0.02 \times \text{organic matter}$$

31 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
32 and stable structures used by kōura as refuge (Usio & Townsend, 2000; NYSTRÖM *et al.*,
1 2006). After stone pile construction, measurements of the size and dimensions of each structure
2 were taken. The post-construction substrate index for stone pile sites was recalculated using
3 each site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
4 and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect
5 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
6 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
7 projected coordinate system (NZTM2000; EPSG:2193).

8 Seasonal variable parameters were recorded at every monitoring round, and included aquatic
9 vegetation cover (emergent and submerged macrophytes) and chemical parameters included
10 water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (µS cm⁻¹) were measured
11 using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured
12 using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

13 Kōura and fish were every monitoring round sampled using two single-winged, double-throated
14 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm;
15 two 50 mm entrance openings) baited with catfish flesh (MacNeil *et al.*, 2025). Fyke nets were
16 placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals,

with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, including eels (*Anguilla* spp.), goldfish (*Carassius auratus* (Linnaeus, 1758)), common smelt (*Retropinna retropinna* (Richardson, 1848)), kōaro (*Galaxias brevipinnis* Günther, 1866), rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792), common bully (*Gobiomorphus cotidianus* McDowall & Fulton, 1978), and mosquitofish (*Gambusia affinis* (S. F. Baird & Girard, 1853)), were recorded for abundance and measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and fish based on the number of nets and traps used at each site for each monitoring round (Kusabs *et al.*, 2015b). Missing kōura weights due to failure of scale ($n = 36$) were estimated using a log –log length–weight regression fitted to individuals with both measurements ($n = 757$), following standard approaches for weight length relationships in aquatic organisms Figure 5. Predicted values were back-transformed with a lognormal bias correction factor ($10^{\{0.5 \cdot \hat{\sigma}^2\}}$) (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.683 \times \log_{10}(\text{OCL length [mm]})$$

6 Data analysis

Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum tests to compare stone pile sites with reference sites. This was done for all response variables and site characteristics to confirm that the stone pile and control sites were equivalent before construction, and to characterise the difference between stone pile and reference sites. We fitted four complementary generalised linear mixed effects models (GLMMs) to assess the effect of construction: a construction-effect model, testing whether kōura or catfish occupancy, abundance, and biomass responded to stone pile construction relative to control sites in a before-after, control-impact (BACI) design (Underwood, 1994); a post-construction trend model, testing whether that response changed progressively over time at stone pile relative to control sites; a convergence model, testing whether stone pile sites approached reference-sites densities over time; and a baseline and equivalence model, testing whether treatment and reference sites were equivalent before construction and after construction.

20 The construction-effect model combined all six post-construction monitoring rounds into one
21 factor level, retaining each round as a separate observation. The occupancy response variable
22 was fitted with a Binomial family with logit link and the abundance and biomass with a Tweedie
23 familie loglink. In the model, period (before/after) and site were fitted as an interaction, with
24 monitoring number and temperature as covariates and site nested within pair as a random
25 effect.

26 The post-construction trend model tested the response variables kōura catfish abundance or
1 biomass fitted with Tweedie, site type and monitoring number as interaction, temperature and
2 substrate index from before construction as covariates and site number as random effect.

3 The convergence model tested the response variables kōura catfish abundance or biomass fitted
4 with Tweedie, site type and monitoring number as interaction, season and substrate index from
5 post-construction as covariates and site number as random effect.

6 The baseline and equivalent models tested the response variables kōura catfish abundance or
7 biomass fitted with Tweedie, site type as fixed effect for the baseline and for the equivalente
8 models the site type and monitoring number as interaction and temperature as covariate and
9 site nested within pair as a random effect.

10 All models were fitted and validated in R (version 4.5.2; R Core Team (2025)) using the
11 glmmTMB, DHARMA, and emmeans packages (Hartig, 2016; Brooks *et al.*, 2017; Lenth
12 & Piaskowski, 2017). Multiple-testing correction using Benjamini–Hochberg correction was
13 applied separately for each model. Robustness of the models was further assessed via sensitivity
14 analyses and colonisation driver comparison in supplementary material.

15 Results

16 Construction-effect model

17 Post-construction trend model

18 Convergence model

19 Figure 2

20 Figure 3

21 Figure 4

22 **Discussion**

23 Most kōura were found along the edges of the stone piles there for recommended future stone
24 piles should be designed to create the most edges.

1 Stone piles were constructed at the shore line to shallow to reach macrophyte beds which
2 are located further offshore. Future stone pile construction could be extended further offshore
3 to reach the macrophyte beds and test if stone piles could suppress macrophyte growth and
4 create pathways for kōura to move between the deeperparts of the lake and the shore.

5 **Conclusion**

6 **Acknowledgements**

7 We thank Te Arawa Lakes Trust and Te Komiti Whakahaere for the opportunity to collect
8 and study the treasured kōura. We are grateful to Soweeta Fort-D'ath and William Anaru
9 (Te Arawa Lakes Trust), and Tihini Grant (Ngāti Pikiao). Also thanks to land owners Piripi
10 Curtis, David McIlraith, Te Takinga Marae, Piki Thomas, Tāheke Opatia Marae, Haddon
1 Lock, Barnett Vercoe, Richard Vercoe, Caroline Newton for access and allowing the build of
2 the build of stone piles. And Martin Sarkezi, Mael Marguet, Ethan Morgan, Logan Kallam
1 and Matt Prentice for assisting in with fieldwork and data collection.

1 **Funding statement**

1 This research was supported by the Fish Futures programme funded through a Ministry of
2 Business, Innovation and Employment grant (CAWX2101) with additional funding provided by
3 the Bay of Plenty Regional Council under the Toihuarewa Waimāori - Bay of Plenty Regional
4 Council Chair in Lake and Freshwater Science programme.

5 **Authors' contributions**

6 Study conception and design were undertaken by O.V.R., I.A.K.K., R.H., F.J.B. and D.Ö.
7 Fieldwork was executed by O.V.R. Data analysis was performed by O.V.R. I.A.K.K obtained
8 council and iwi consent and provided local ecological knowledge. Manuscript drafting was led

9 by O.V.R. All authors contributed to manuscript revision, approved the results, and the final
1 manuscript.

2 Use of Artificial Intelligence

3 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
4 language refinement. All content was critically reviewed and the authors take full responsibility
5 for the final manuscript.

6 Declarations

7 The corresponding author has declared that none of the authors has any competing interests.

8 Data availability statement

9 The code and derived data supporting the findings of this study are openly available on GitHub
10 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
11 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
12 archived on Zenodo (<https://doi.org/.....>). Peer review documents, including the cover letter
13 and reviewer responses, are available in the manuscript repository in the interest of open and
14 transparent science.

15 Tables

Table 1: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test, pairs matched by pair_id, supporting the BACI design) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test, documenting the starting gap relative to natural conditions, motivating the convergence analysis). With only 5 paired sites, these are low-power checks — a non-significant result indicates no obvious gross imbalance, not formal proof of equivalence.

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Catfish CPUE	0.05 ± 0.11	0.1 ± 0.22	0.3 ± 0.52	1.000	0.420
Kōura BPUE	7.95 ± 12.15	2.8 ± 3.84	36.69 ± 36.82	0.789	0.206
Catfish BPUE	0.15 ± 0.34	24.6 ± 55.01	97.28 ± 232.15	1.000	0.343
Substrate index (before)	3.39 ± 1.07	3.68 ± 0.95	5.29 ± 1.05	0.789	0.012
Substrate index (after)	3.65 ± 1.12	3.68 ± 0.95	5.29 ± 1.05	1.000	0.032
Slope (0–20 m)	0.06 ± 0.05	0.06 ± 0.04	0.09 ± 0.05	0.590	0.245
Temperature (°C)	24.74 ± 1.92	24.78 ± 1.47	24.48 ± 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 ± 1.48	9.63 ± 1.29	10.03 ± 0.86	0.418	0.426
Specific conductivity (µS/cm)	195.44 ± 70.05	199.8 ± 78.52	165.45 ± 8.5	0.281	0.951
pH	7.5 ± 2.15	7.61 ± 2.2	8.53 ± 0.66	1.000	0.426

¹⁶ Source: [Article Notebook](#)

¹⁷ Figures

¹⁸ Supplementary information

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re- sponse	com- pari- son	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- eff	BLow	high
M1okRQ1Kōura	occu- pancy	vs_con- trol	full	(Intercept)	-	4.736	-	0.090	NA	- 1.249
					8.033	1.696			17.315	

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-species	com- pari- response_type	time_frame	es- ti- mate	std.er-	statis- tic	p.value	adjco- B	low	high
M1okRQ1Kōura	occu- pancy	vs_confull trol	PeriodAfter	4.171	2.500	1.669	0.095	NA	-	9.071
									0.728	
M1okRQ1Kōura	occu- pancy	vs_confull trol	siteStone_pile	0.017	2.482	0.007	0.995	NA	-	4.883
									4.849	
M1okRQ1Kōura	occu- pancy	vs_confull trol	monitoring_int	-	0.333	-	0.512	NA	-	0.435
				0.218		0.655			0.871	
M1okRQ1Kōura	occu- pancy	vs_confull trol	Temperature	0.292	0.171	1.713	0.087	NA	-	0.626
									0.042	
M1okRQ1Kōura	occu- pancy	vs_confull trol	Peri- odAfter:site- Stone_pile	1.814	1.968	0.922	0.357	0.853	-	5.671
									2.043	
M1akRQ1Kōura	bun- dance	vs_confull trol	(Intercept)	-	1.553	-	0.327	NA	-	1.523
				1.521		0.979			4.566	
M1akRQ1Kōura	bun- dance	vs_confull trol	PeriodAfter	1.659	1.043	1.591	0.112	NA	-	3.703
									0.384	
M1akRQ1Kōura	bun- dance	vs_confull trol	siteStone_pile	0.702	1.306	0.538	0.591	NA	-	3.261
									1.857	
M1akRQ1Kōura	bun- dance	vs_confull trol	monitoring_int	-	0.096	-	0.349	NA	-	0.099
				0.090		0.937			0.279	
M1akRQ1Kōura	bun- dance	vs_confull trol	Temperature	-	0.042	-	0.235	NA	-	0.032
				0.050		1.189			0.132	
M1akRQ1Kōura	bun- dance	vs_confull trol	Peri- odAfter:site- Stone_pile	0.283	1.214	0.233	0.816	0.853	-	2.661
									2.096	
M1bkRQ1Kōura	biomass	vs_confull trol	(Intercept)	0.883	1.951	0.452	0.651	NA	-	4.706
									2.941	
M1bkRQ1Kōura	biomass	vs_confull trol	PeriodAfter	1.786	1.374	1.300	0.194	NA	-	4.479
									0.907	

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-species	com- pari- response_type	time_frame	es- ti- mate	std. ror	er- statis- tic	p.value	adjco- B	low	high
M1bkRQ1Kōura	biomass	vs_confull	siteStone_pile	1.370	1.550	0.884	0.377	NA	-	4.407
		trol							1.667	
M1bkRQ1Kōura	biomass	vs_confull	monitoring_int	-	0.133	-	0.751	NA	-	0.219
		trol		0.042		0.317			0.304	
M1bkRQ1Kōura	biomass	vs_confull	Temperature	-	0.058	-	0.708	NA	-	0.091
		trol		0.022		0.374			0.135	
M1bkRQ1Kōura	biomass	vs_confull	Peri-	-	1.487	-	0.774	0.853	-	2.488
		trol	odAfter:site-	0.426		0.287			3.341	
			Stone_pile							
M1locRQ1Cat-	occu-	vs_confull	(Intercept)	-	4.271	-	0.150	NA	-	2.225
fish	pancy	trol		6.146		1.439			14.517	
M1locRQ1Cat-	occu-	vs_confull	PeriodAfter	0.402	2.336	0.172	0.863	NA	-	4.980
fish	pancy	trol							4.176	
M1locRQ1Cat-	occu-	vs_confull	siteStone_pile	0.005	2.095	0.002	0.998	NA	-	4.110
fish	pancy	trol							4.101	
M1locRQ1Cat-	occu-	vs_confull	monitoring_int	0.112	0.343	0.325	0.745	NA	-	0.784
fish	pancy	trol							0.561	
M1locRQ1Cat-	occu-	vs_confull	Temperature	0.160	0.147	1.091	0.275	NA	-	0.449
fish	pancy	trol							0.128	
M1locRQ1Cat-	occu-	vs_confull	Peri-	0.426	2.302	0.185	0.853	0.853	-	4.939
fish	pancy	trol	odAfter:site-						4.087	
			Stone_pile							
M1lacRQ1Cat-	abun-	vs_confull	(Intercept)	-	NA	NA	NA	NA	NA	NA
fish	dance	trol		9.453						
M1lacRQ1Cat-	abun-	vs_confull	PeriodAfter	2.569	NA	NA	NA	NA	NA	NA
fish	dance	trol								
M1lacRQ1Cat-	abun-	vs_confull	siteStone_pile	-	NA	NA	NA	NA	NA	NA
fish	dance	trol		0.721						

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	RQ	species	re-	com- pari- response_type	time_frame	es- ti- mate	std.er-	statis- tic	p.value	up_adjco	BLow	conf.high
M1ac	RQ1	Cat-	abun-	vs_confull	monitoring_int	-	NA	NA	NA	NA	NA	NA
		fish	dance	trol		0.422						
M1ac	RQ1	Cat-	abun-	vs_confull	Temperature	0.233	NA	NA	NA	NA	NA	NA
		fish	dance	trol								
M1ac	RQ1	Cat-	abun-	vs_confull	Peri-	1.872	NA	NA	NA	NA	NA	NA
		fish	dance	trol	odAfter:site-							
					Stone_pile							
M1ac	RQ1	Cat-	abun-	vs_confull	(Intercept)	-	NA	NA	NA	NA	NA	NA
		fish	dance	trol		1.472						
M1bc	RQ1	Cat-	biomass	vs_confull	(Intercept)	0.544	6.130	0.089	0.929	NA	-	12.558
		fish		trol							11.470	
M1bc	RQ1	Cat-	biomass	vs_confull	PeriodAfter	-	3.271	-	0.899	NA	-	5.997
		fish		trol		0.414		0.127			6.824	
M1bc	RQ1	Cat-	biomass	vs_confull	siteStone_pile	-	5.616	-	0.352	NA	-	5.778
		fish		trol		5.230		0.931			16.238	
M1bc	RQ1	Cat-	biomass	vs_confull	monitoring_int	-	0.631	-	0.842	NA	-	1.111
		fish		trol		0.125		0.199			1.362	
M1bc	RQ1	Cat-	biomass	vs_confull	Temperature	0.042	0.212	0.199	0.842	NA	-	0.457
		fish		trol							0.373	
M1bc	RQ1	Cat-	biomass	vs_confull	Peri-	4.954	5.473	0.905	0.365	0.853	-	15.682
		fish		trol	odAfter:site-						5.774	
					Stone_pile							
M2ak	RQ2	Kōura	abun-	vs_conpost	(Intercept)	-	2.418	-	0.350	NA	-	2.480
			dance	trol		2.259		0.934			6.998	
M2ak	RQ2	Kōura	abun-	vs_conpost	siteStone_pile	0.457	1.196	0.382	0.702	NA	-	2.800
			dance	trol							1.886	
M2ak	RQ2	Kōura	abun-	vs_conpost	monitoring_int	-	0.164	-	0.074	NA	-	0.028
			dance	trol		0.294		1.790			0.616	

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100	Q101	Q102	Q103	Q104	Q105	Q106	Q107	Q108	Q109	Q110	Q111	Q112	Q113	Q114	Q115	Q116	Q117	Q118	Q119	Q120	Q121	Q122	Q123	Q124	Q125	Q126	Q127	Q128	Q129	Q130	Q131	Q132	Q133	Q134	Q135	Q136	Q137	Q138	Q139	Q140	Q141	Q142	Q143	Q144	Q145	Q146	Q147	Q148	Q149	Q150	Q151	Q152	Q153	Q154	Q155	Q156	Q157	Q158	Q159	Q160	Q161	Q162	Q163	Q164	Q165	Q166	Q167	Q168	Q169	Q170	Q171	Q172	Q173	Q174	Q175	Q176	Q177	Q178	Q179	Q180	Q181	Q182	Q183	Q184	Q185	Q186	Q187	Q188	Q189	Q190	Q191	Q192	Q193	Q194	Q195	Q196	Q197	Q198	Q199	Q200	Q201	Q202	Q203	Q204	Q205	Q206	Q207	Q208	Q209	Q210	Q211	Q212	Q213	Q214	Q215	Q216	Q217	Q218	Q219	Q220	Q221	Q222	Q223	Q224	Q225	Q226	Q227	Q228	Q229	Q230	Q231	Q232	Q233	Q234	Q235	Q236	Q237	Q238	Q239	Q240	Q241	Q242	Q243	Q244	Q245	Q246	Q247	Q248	Q249	Q250	Q251	Q252	Q253	Q254	Q255	Q256	Q257	Q258	Q259	Q260	Q261	Q262	Q263	Q264	Q265	Q266	Q267	Q268	Q269	Q270	Q271	Q272	Q273	Q274	Q275	Q276	Q277	Q278	Q279	Q280	Q281	Q282	Q283	Q284	Q285	Q286	Q287	Q288	Q289	Q290	Q291	Q292	Q293	Q294	Q295	Q296	Q297	Q298	Q299	Q300	Q301	Q302	Q303	Q304	Q305	Q306	Q307	Q308	Q309	Q310	Q311	Q312	Q313	Q314	Q315	Q316	Q317	Q318	Q319	Q320	Q321	Q322	Q323	Q324	Q325	Q326	Q327	Q328	Q329	Q330	Q331	Q332	Q333	Q334	Q335	Q336	Q337	Q338	Q339	Q340	Q341	Q342	Q343	Q344	Q345	Q346	Q347	Q348	Q349	Q350	Q351	Q352	Q353	Q354	Q355	Q356	Q357	Q358	Q359	Q360	Q361	Q362	Q363	Q364	Q365	Q366	Q367	Q368	Q369	Q370	Q371	Q372	Q373	Q374	Q375	Q376	Q377	Q378	Q379	Q380	Q381	Q382	Q383	Q384	Q385	Q386	Q387	Q388	Q389	Q390	Q391	Q392	Q393	Q394	Q395	Q396	Q397	Q398	Q399	Q400	Q401	Q402	Q403	Q404	Q405	Q406	Q407	Q408	Q409	Q410	Q411	Q412	Q413	Q414	Q415	Q416	Q417	Q418	Q419	Q420	Q421	Q422	Q423	Q424	Q425	Q426	Q427	Q428	Q429	Q430	Q431	Q432	Q433	Q434	Q435	Q436	Q437	Q438	Q439	Q440	Q441	Q442	Q443	Q444	Q445	Q446	Q447	Q448	Q449	Q450	Q451	Q452	Q453	Q454	Q455	Q456	Q457	Q458	Q459	Q460	Q461	Q462	Q463	Q464	Q465	Q466	Q467	Q468	Q469	Q470	Q471	Q472	Q473	Q474	Q475	Q476	Q477	Q478	Q479	Q480	Q481	Q482	Q483	Q484	Q485	Q486	Q487	Q488	Q489	Q490	Q491	Q492	Q493	Q494	Q495	Q496	Q497	Q498	Q499	Q500	Q501	Q502	Q503	Q504	Q505	Q506	Q507	Q508	Q509	Q510	Q511	Q512	Q513	Q514	Q515	Q516	Q517	Q518	Q519	Q520	Q521	Q522	Q523	Q524	Q525	Q526	Q527	Q528	Q529	Q530	Q531	Q532	Q533	Q534	Q535	Q536	Q537	Q538	Q539	Q540	Q541	Q542	Q543	Q544	Q545	Q546	Q547	Q548	Q549	Q550	Q551	Q552	Q553	Q554	Q555	Q556	Q557	Q558	Q559	Q560	Q561	Q562	Q563	Q564	Q565	Q566	Q567	Q568	Q569	Q570	Q571	Q572	Q573	Q574	Q575	Q576	Q577	Q578	Q579	Q580	Q581	Q582	Q583	Q584	Q585	Q586	Q587	Q588	Q589	Q590	Q591	Q592	Q593	Q594	Q595	Q596	Q597	Q598	Q599	Q600	Q601	Q602	Q603	Q604	Q605	Q606	Q607	Q608	Q609	Q610	Q611	Q612	Q613	Q614	Q615	Q616	Q617	Q618	Q619	Q620	Q621	Q622	Q623	Q624	Q625	Q626	Q627	Q628	Q629	Q630	Q631	Q632	Q633	Q634	Q635	Q636	Q637	Q638	Q639	Q640	Q641	Q642	Q643	Q644	Q645	Q646	Q647	Q648	Q649	Q650	Q651	Q652	Q653	Q654	Q655	Q656	Q657	Q658	Q659	Q660	Q661	Q662	Q663	Q664	Q665	Q666	Q667	Q668	Q669	Q670	Q671	Q672	Q673	Q674	Q675	Q676	Q677	Q678	Q679	Q680	Q681	Q682	Q683	Q684	Q685	Q686	Q687	Q688	Q689	Q690	Q691	Q692	Q693	Q694	Q695	Q696	Q697	Q698	Q699	Q700	Q701	Q702	Q703	Q704	Q705	Q706	Q707	Q708	Q709	Q710	Q711	Q712	Q713	Q714	Q715	Q716	Q717	Q718	Q719	Q720	Q721	Q722	Q723	Q724	Q725	Q726	Q727	Q728	Q729	Q730	Q731	Q732	Q733	Q734	Q735	Q736	Q737	Q738	Q739	Q740	Q741	Q742	Q743	Q744	Q745	Q746	Q747	Q748	Q749	Q750	Q751	Q752	Q753	Q754	Q755	Q756	Q757	Q758	Q759	Q760	Q761	Q762	Q763	Q764	Q765	Q766	Q767	Q768	Q769	Q770	Q771	Q772	Q773	Q774	Q775	Q776	Q777	Q778	Q779	Q780	Q781	Q782	Q783	Q784	Q785	Q786	Q787	Q788	Q789	Q790	Q791	Q792	Q793	Q794	Q795	Q796	Q797	Q798	Q799	Q800	Q801	Q802	Q803	Q804	Q805	Q806	Q807	Q808	Q809	Q810	Q811	Q812	Q813	Q814	Q815	Q816	Q817	Q818	Q819	Q820	Q821	Q822	Q823	Q824	Q825	Q826	Q827	Q828	Q829	Q830	Q831	Q832	Q833	Q834	Q835	Q836	Q837	Q838	Q839	Q840	Q841	Q842	Q843	Q844	Q845	Q846	Q847	Q848	Q849	Q850	Q851	Q852	Q853	Q854	Q855	Q856	Q857	Q858	Q859	Q860	Q861	Q862	Q863	Q864	Q865	Q866	Q867	Q868	Q869	Q870	Q871	Q872	Q873	Q874	Q875	Q876	Q877	Q878	Q879	Q880	Q881	Q882	Q883	Q884	Q885	Q886	Q887	Q888	Q889	Q890	Q891	Q892	Q893	Q894	Q895	Q896	Q897	Q898	Q899	Q900	Q901	Q902	Q903	Q904	Q905	Q906	Q907	Q908	Q909	Q910	Q911	Q912	Q913	Q914	Q915	Q916	Q917	Q918	Q919	Q920	Q921	Q922	Q923	Q924	Q925	Q926	Q927	Q928	Q929	Q930	Q931	Q932	Q933	Q934	Q935	Q936	Q937	Q938	Q939	Q940	Q941	Q942	Q943	Q944	Q945	Q946	Q947	Q948	Q949	Q950	Q951	Q952	Q953	Q954	Q955	Q956	Q957	Q958	Q959	Q960	Q961	Q962	Q963	Q964	Q965	Q966	Q967	Q968	Q969	Q970	Q971	Q972	Q973	Q974	Q975	Q976	Q977	Q978	Q979	Q980	Q981	Q982	Q983	Q984	Q985	Q986	Q987	Q988	Q989	Q990	Q991	Q992	Q993	Q994	Q995	Q996	Q997	Q998	Q999	Q1000
re-	com-	pari-	time	frame	es-	std.er-	statis-	p.value	adjco	BH	low	high																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
M2akRQ2Kōura	abun-	vs_conpost	Temperature	-	0.043	-	0.205	NA	-	0.030																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	dance	trol		0.055	1.267		0.140																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2akRQ2Kōura	abun-	vs_conpost	Substrate_in-	0.785	0.563	1.395	0.163	NA	-	1.888																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	dance	trol	dex_before						0.318																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
M2akRQ2Kōura	abun-	vs_conpost	site-	0.322	0.202	1.593	0.111	0.148	-	0.718																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	dance	trol	Stone_pile:mon-						0.074																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M2bkRQ2Kōura	biomass	vs_conpost	(Intercept)	1.128	2.305	0.489	0.625	NA	-	5.645																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol								3.389																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
M2bkRQ2Kōura	biomass	vs_conpost	siteStone_pile	-	1.215	-	0.991	NA	-	2.367																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol			0.014	0.012		2.396																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2bkRQ2Kōura	biomass	vs_conpost	monitoring_int	-	0.228	-	0.157	NA	-	0.124																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol			0.323	1.417		0.769																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2bkRQ2Kōura	biomass	vs_conpost	Temperature	-	0.060	-	0.685	NA	-	0.093																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol			0.024	0.405		0.142																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2bkRQ2Kōura	biomass	vs_conpost	Substrate_in-	0.607	0.494	1.230	0.219	NA	-	1.576																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol		dex_before						0.361																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
M2bkRQ2Kōura	biomass	vs_conpost	site-	0.448	0.280	1.601	0.109	0.148	-	0.996																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	trol		Stone_pile:mon-						0.100																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M2acRQ2Cat-	abun-	vs_conpost	(Intercept)	-	4.312	-	0.643	NA	-	6.456																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	fish	dance		1.996	0.463		10.448																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2acRQ2Cat-	abun-	vs_conpost	siteStone_pile	-	2.402	-	0.123	NA	-	1.003																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	fish	dance		3.705	1.542		8.413																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
M2acRQ2Cat-	abun-	vs_conpost	monitoring_int	-	0.385	-	0.029	NA	-	-																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	fish	dance		0.843	2.188		1.598	0.088																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
M2acRQ2Cat-	abun-	vs_conpost	Temperature	0.244	0.085	2.875	0.004	NA	0.078	0.410																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100	Q101	Q102	Q103	Q104	Q105	Q106	Q107	Q108	Q109	Q110	Q111	Q112	Q113	Q114	Q115	Q116	Q117	Q118	Q119	Q120	Q121	Q122	Q123	Q124	Q125	Q126	Q127	Q128	Q129	Q130	Q131	Q132	Q133	Q134	Q135	Q136	Q137	Q138	Q139	Q140	Q141	Q142	Q143	Q144	Q145	Q146	Q147	Q148	Q149	Q150	Q151	Q152	Q153	Q154	Q155	Q156	Q157	Q158	Q159	Q160	Q161	Q162	Q163	Q164	Q165	Q166	Q167	Q168	Q169	Q170	Q171	Q172	Q173	Q174	Q175	Q176	Q177	Q178	Q179	Q180	Q181	Q182	Q183	Q184	Q185	Q186	Q187	Q188	Q189	Q190	Q191	Q192	Q193	Q194	Q195	Q196	Q197	Q198	Q199	Q200	Q201	Q202	Q203	Q204	Q205	Q206	Q207	Q208	Q209	Q210	Q211	Q212	Q213	Q214	Q215	Q216	Q217	Q218	Q219	Q220	Q221	Q222	Q223	Q224	Q225	Q226	Q227	Q228	Q229	Q230	Q231	Q232	Q233	Q234	Q235	Q236	Q237	Q238	Q239	Q240	Q241	Q242	Q243	Q244	Q245	Q246	Q247	Q248	Q249	Q250	Q251	Q252	Q253	Q254	Q255	Q256	Q257	Q258	Q259	Q260	Q261	Q262	Q263	Q264	Q265	Q266	Q267	Q268	Q269	Q270	Q271	Q272	Q273	Q274	Q275	Q276	Q277	Q278	Q279	Q280	Q281	Q282	Q283	Q284	Q285	Q286	Q287	Q288	Q289	Q290	Q291	Q292	Q293	Q294	Q295	Q296	Q297	Q298	Q299	Q300	Q301	Q302	Q303	Q304	Q305	Q306	Q307	Q308	Q309	Q310	Q311	Q312	Q313	Q314	Q315	Q316	Q317	Q318	Q319	Q320	Q321	Q322	Q323	Q324	Q325	Q326	Q327	Q328	Q329	Q330	Q331	Q332	Q333	Q334	Q335	Q336	Q337	Q338	Q339	Q340	Q341	Q342	Q343	Q344	Q345	Q346	Q347	Q348	Q349	Q350	Q351	Q352	Q353	Q354	Q355	Q356	Q357	Q358	Q359	Q360	Q361	Q362	Q363	Q364	Q365	Q366	Q367	Q368	Q369	Q370	Q371	Q372	Q373	Q374	Q375	Q376	Q377	Q378	Q379	Q380	Q381	Q382	Q383	Q384	Q385	Q386	Q387	Q388	Q389	Q390	Q391	Q392	Q393	Q394	Q395	Q396	Q397	Q398	Q399	Q400	Q401	Q402	Q403	Q404	Q405	Q406	Q407	Q408	Q409	Q410	Q411	Q412	Q413	Q414	Q415	Q416	Q417	Q418	Q419	Q420	Q421	Q422	Q423	Q424	Q425	Q426	Q427	Q428	Q429	Q430	Q431	Q432	Q433	Q434	Q435	Q436	Q437	Q438	Q439	Q440	Q441	Q442	Q443	Q444	Q445	Q446	Q447	Q448	Q449	Q450	Q451	Q452	Q453	Q454	Q455	Q456	Q457	Q458	Q459	Q460	Q461	Q462	Q463	Q464	Q465	Q466	Q467	Q468	Q469	Q470	Q471	Q472	Q473	Q474	Q475	Q476	Q477	Q478	Q479	Q480	Q481	Q482	Q483	Q484	Q485	Q486	Q487	Q488	Q489	Q490	Q491	Q492	Q493	Q494	Q495	Q496	Q497	Q498	Q499	Q500	Q501	Q502	Q503	Q504	Q505	Q506	Q507	Q508	Q509	Q510	Q511	Q512	Q513	Q514	Q515	Q516	Q517	Q518	Q519	Q520	Q521	Q522	Q523	Q524	Q525	Q526	Q527	Q528	Q529	Q530	Q531	Q532	Q533	Q534	Q535	Q536	Q537	Q538	Q539	Q540	Q541	Q542	Q543	Q544	Q545	Q546	Q547	Q548	Q549	Q550	Q551	Q552	Q553	Q554	Q555	Q556	Q557	Q558	Q559	Q560	Q561	Q562	Q563	Q564	Q565	Q566	Q567	Q568	Q569	Q570	Q571	Q572	Q573	Q574	Q575	Q576	Q577	Q578	Q579	Q580	Q581	Q582	Q583	Q584	Q585	Q586	Q587	Q588	Q589	Q590	Q591	Q592	Q593	Q594	Q595	Q596	Q597	Q598	Q599	Q600	Q601	Q602	Q603	Q604	Q605	Q606	Q607	Q608	Q609	Q610	Q611	Q612	Q613	Q614	Q615	Q616	Q617	Q618	Q619	Q620	Q621	Q622	Q623	Q624	Q625	Q626	Q627	Q628	Q629	Q630	Q631	Q632	Q633	Q634	Q635	Q636	Q637	Q638	Q639	Q640	Q641	Q642	Q643	Q644	Q645	Q646	Q647	Q648	Q649	Q650	Q651	Q652	Q653	Q654	Q655	Q656	Q657	Q658	Q659	Q660	Q661	Q662	Q663	Q664	Q665	Q666	Q667	Q668	Q669	Q670	Q671	Q672	Q673	Q674	Q675	Q676	Q677	Q678	Q679	Q680	Q681	Q682	Q683	Q684	Q685	Q686	Q687	Q688	Q689	Q690	Q691	Q692	Q693	Q694	Q695	Q696	Q697	Q698	Q699	Q700	Q701	Q702	Q703	Q704	Q705	Q706	Q707	Q708	Q709	Q710	Q711	Q712	Q713	Q714	Q715	Q716	Q717	Q718	Q719	Q720	Q721	Q722	Q723	Q724	Q725	Q726	Q727	Q728	Q729	Q730	Q731	Q732	Q733	Q734	Q735	Q736	Q737	Q738	Q739	Q740	Q741	Q742	Q743	Q744	Q745	Q746	Q747	Q748	Q749	Q750	Q751	Q752	Q753	Q754	Q755	Q756	Q757	Q758	Q759	Q760	Q761	Q762	Q763	Q764	Q765	Q766	Q767	Q768	Q769	Q770	Q771	Q772	Q773	Q774	Q775	Q776	Q777	Q778	Q779	Q780	Q781	Q782	Q783	Q784	Q785	Q786	Q787	Q788	Q789	Q790	Q791	Q792	Q793	Q794	Q795	Q796	Q797	Q798	Q799	Q800	Q801	Q802	Q803	Q804	Q805	Q806	Q807	Q808	Q809	Q810	Q811	Q812	Q813	Q814	Q815	Q816	Q817	Q818	Q819	Q820	Q821	Q822	Q823	Q824	Q825	Q826	Q827	Q828	Q829	Q830	Q831	Q832	Q833	Q834	Q835	Q836	Q837	Q838	Q839	Q840	Q841	Q842	Q843	Q844	Q845	Q846	Q847	Q848	Q849	Q850	Q851	Q852	Q853	Q854	Q855	Q856	Q857	Q858	Q859	Q860	Q861	Q862	Q863	Q864	Q865	Q866	Q867	Q868	Q869	Q870	Q871	Q872	Q873	Q874	Q875	Q876	Q877	Q878	Q879	Q880	Q881	Q882	Q883	Q884	Q885	Q886	Q887	Q888	Q889	Q890	Q891	Q892	Q893	Q894	Q895	Q896	Q897	Q898	Q899	Q900	Q901	Q902	Q903	Q904	Q905	Q906	Q907	Q908	Q909	Q910	Q911	Q912	Q913	Q914	Q915	Q916	Q917	Q918	Q919	Q920	Q921	Q922	Q923	Q924	Q925	Q926	Q927	Q928	Q929	Q930	Q931	Q932	Q933	Q934	Q935	Q936	Q937	Q938	Q939	Q940	Q941	Q942	Q943	Q944	Q945	Q946	Q947	Q948	Q949	Q950	Q951	Q952	Q953	Q954	Q955	Q956	Q957	Q958	Q959	Q960	Q961	Q962	Q963	Q964	Q965	Q966	Q967	Q968	Q969	Q970	Q971	Q972	Q973	Q974	Q975	Q976	Q977	Q978	Q979	Q980	Q981	Q982	Q983	Q984	Q985	Q986	Q987	Q988	Q989	Q990	Q991	Q992	Q993	Q994	Q995	Q996	Q997	Q998	Q999	Q1000
re-	com-	es-	ti-	std.er-	statis-	p.value	adjco	BL	low	high																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
M3akRQ3Kōura	abun-	vs_ref-full	monitoring_int	-	0.060	-	0.093	NA	-	0.017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-			0.101	1.680		0.219																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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M3akRQ3Kōura	abun-	vs_ref-full	SeasonSpring	-	0.262	-	0.210	NA	-	0.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-			0.328	1.255		0.842																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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M3akRQ3Kōura	abun-	vs_ref-full	SeasonSummer	-	0.214	-	0.031	NA	-	-																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-			0.463	2.162		0.882	0.043																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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M3akRQ3Kōura	abun-	vs_ref-full	SeasonWinter	-	0.264	-	0.113	NA	-	0.099																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-			0.419	1.587		0.937																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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M3akRQ3Kōura	abun-	vs_ref-full	Substrate_in-	0.626	0.420	1.489	0.137	NA	-	1.449																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-		dex_after						0.198																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M3akRQ3Kōura	abun-	vs_ref-full	site-	0.203	0.115	1.761	0.078	0.114	-	0.430																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
dance	er-		Stone_pile:mon-						0.023																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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M3bkRQ3Kōura	biomass	vs_ref-full	(Intercept)	1.427	2.125	0.671	0.502	NA	-	5.592																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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M3bkRQ3Kōura	biomass	vs_ref-full	siteStone_pile	-	1.091	-	0.923	NA	-	2.033																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	er-			0.105	0.096		2.243																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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M3bkRQ3Kōura	biomass	vs_ref-full	monitoring_int	-	0.075	-	0.249	NA	-	0.061																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	er-			0.087	1.153		0.234																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re-species	com- pari- response_type	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- BIC	low	high
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonSpring	- 0.342	- 0.127	NA	- 0.148		
		er- ence		0.522	1.528			1.193		
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonSummer	- 0.262	- 0.109	NA	- 0.094		
		er- ence		0.420	1.603			0.934		
M3bkRQ3	Kōura	biomass	vs_ref-full	SeasonWinter	- 0.334	- 0.153	NA	- 0.177		
		er- ence		0.477	1.430			1.132		
M3bkRQ3	Kōura	biomass	vs_ref-full	Substrate_in- dex_after	0.406	0.380	1.067	0.286	NA	- 1.150
		er- ence						0.339		
M3bkRQ3	Kōura	biomass	vs_ref-full	site- Stone_pile:mon- itoring_int	0.239	0.139	1.718	0.086	0.114	- 0.512
		er- ence						0.034		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	(Intercept)	- 3.698	- 0.124	NA	- 1.552		
		er- ence		5.696	1.540			12.944		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	siteStone_pile	- 2.125	- 0.615	NA	- 3.098		
		er- ence		1.067	0.502			5.233		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	monitoring_int	0.159	0.091	1.747	0.081	NA	- 0.338
		er- ence						0.019		
M3acRQ3	Cat- fish	abun- dance	vs_ref-full	SeasonSpring	- 0.547	- 0.852	NA	- 0.970		
		er- ence		0.102	0.187			1.174		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	species	response	com- pari- type	time_frame	es- ti- mate	std.er	statis- tic	p.value	adjco	BLow	HI	conf.high
M3ac	RQ3	Cat-	abun-	vs_ref-full	SeasonSummer		1.321	0.359	3.685	0.000	NA	0.618	2.024	
		fish	dance	er- ence										
M3ac	RQ3	Cat-	abun-	vs_ref-full	SeasonWinter		-	0.649	-	0.534	NA	-	0.869	
		fish	dance	er- ence			0.404		0.622			1.676		
M3ac	RQ3	Cat-	abun-	vs_ref-full	Substrate_in-		0.486	0.662	0.734	0.463	NA	-	1.785	
		fish	dance	er- ence	dex_after							0.812		
M3ac	RQ3	Cat-	abun-	vs_ref-full	site-		0.340	0.263	1.293	0.196	0.196	-	0.854	
		fish	dance	er- ence	Stone_pile:mon-							0.175		
					itoring_int									
M3ac	RQ3	Cat-	abun-	vs_ref-full	(Intercept)		-	10860.696	-	0.998	NA	-	21265.816	
		fish	dance	er- ence			20.757		0.002			21307.331		
M3bc	RQ3	Cat-	biomass	vs_ref-full	(Intercept)		-	5.884	-	0.789	NA	-	9.959	
		fish		er- ence			1.573		0.267			13.106		
M3bc	RQ3	Cat-	biomass	vs_ref-full	siteStone_pile		-	4.013	-	0.163	NA	-	2.270	
		fish		er- ence			5.596		1.394			13.462		
M3bc	RQ3	Cat-	biomass	vs_ref-full	monitoring_int		-	0.178	-	0.957	NA	-	0.339	
		fish		er- ence			0.010		0.054			0.358		
M3bc	RQ3	Cat-	biomass	vs_ref-full	SeasonSpring		-	0.919	-	0.608	NA	-	1.329	
		fish		er- ence			0.472		0.513			2.273		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	species	re- response	com- pari- type	time	frame	es- mate	std. ror	er- tic	stat- p.value	adjco- B	How	conf.high
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full	SeasonSummer	0.878	0.664	1.323	0.186	NA	-	2.178
											0.423		
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full	SeasonWinter	0.207	0.861	0.240	0.810	NA	-	1.894
											1.480		
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full	Substrate_in- dex_after	0.571	1.054	0.541	0.588	NA	-	2.637
											1.495		
M3bcRQ3	Cat-	fish	biomass	svs_ref- er- ence	full	site- Stone_pile:mon- itoring_int	1.223	0.642	1.905	0.057	0.114	-	2.481
											0.035		
M4TBQ4	Kōura	abun-	treat-	be-		(Intercept)	-	0.785	-	0.016	NA	-	-
		dance	ment	fore			1.897		2.415		3.437	0.358	
M4TBQ4	Kōura	abun-	treat-	be-		Site_type_de-	0.288	1.044	0.276	0.783	0.783	-	2.334
		dance	ment	fore		tailStone_pile					1.759		
M4TBQ4	Kōura	biomass	treat-	be-		(Intercept)	1.030	0.830	1.241	0.215	NA	-	2.656
			ment	fore							0.597		
M4TBQ4	Kōura	biomass	treat-	be-		Site_type_de-	1.044	0.977	1.068	0.286	0.571	-	2.959
			ment	fore		tailStone_pile					0.872		
M4TBQ4	Cat-	fish	abun-	treat-	be-	(Intercept)	-	0.908	-	0.011	NA	-	-
		dance	ment	fore			2.303		2.535		4.083	0.522	
M4TBQ4	Cat-	fish	abun-	treat-	be-	Site_type_de-	-	1.535	-	0.652	0.783	-	2.315
		dance	ment	fore		tailStone_pile	0.693		0.452		3.701		
M4TBQ4	Cat-	fish	biomass	treat-	be-	(Intercept)	3.203	1.279	2.504	0.012	NA	0.696	5.709
			ment	fore									
M4TBQ4	Cat-	fish	biomass	treat-	be-	Site_type_de-	-	3.107	-	0.101	0.403	-	0.990
			ment	fore		tailStone_pile	5.100		1.641		11.190		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re- Q	com- pari- response_type	time_frame	es- ti- std.er-stat- tic	p.value	adjco	BF	low	high
M4TABQ4	Kōura	abun-	treat-	af-	monitoring_int	- 0.165	- 0.069	NA	- 0.023
		dance	ment	ter		0.299	1.820	0.622	
M4TABQ4	Kōura	abun-	treat-	af-	Temperature	- 0.044	- 0.220	NA	- 0.032
		dance	ment	ter		0.053	1.226	0.139	
M4TABQ4	Kōura	abun-	treat-	af-	Site_type_de-	0.328	0.203	1.619	0.105
		dance	ment	ter	tail-			0.112	- 0.726
					Stone_pile:mon-			0.069	
					itoring_int				
M4TABQ4	Kōura	biomas	treat-	af-	(Intercept)	3.154	0.872	3.618	0.000
		ment	ter					NA	1.445
								4.863	
M4TABQ4	Kōura	biomas	treat-	af-	Site_type_de-	- 0.996	- 0.704	NA	- 1.574
		ment	ter		tailStone_pile	0.378	0.379	2.329	
M4TABQ4	Kōura	biomas	treat-	af-	monitoring_int	- 0.220	- 0.117	NA	- 0.086
		ment	ter			0.345	1.568	0.775	
M4TABQ4	Kōura	biomas	treat-	af-	Site_type_de-	0.459	0.276	1.661	0.097
		ment	ter		tail-			0.112	- 1.000
					Stone_pile:mon-			0.083	
					itoring_int				
M4TABQ4	Cat-	abun-	treat-	af-	(Intercept)	- 1.629	- 0.169	NA	- 0.950
	fish	dance	ment	ter		2.243	1.377	5.435	
M4TABQ4	Cat-	abun-	treat-	af-	Site_type_de-	- 1.839	- 0.150	NA	- 0.959
	fish	dance	ment	ter	tailStone_pile	2.645	1.439	6.248	
M4TABQ4	Cat-	abun-	treat-	af-	monitoring_int	- 0.458	- 0.201	NA	- 0.313
	fish	dance	ment	ter		0.586	1.278	1.484	
M4TABQ4	Cat-	abun-	treat-	af-	Site_type_de-	1.166	0.586	1.990	0.047
	fish	dance	ment	ter	tail-			0.112	0.018
					Stone_pile:mon-			2.315	
					itoring_int				

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Re- Q	species	com- pari- response_type	time_frame	term	es- mate	std.er	statis- tic	p.value	adjco- B	How	conf.high
M4TABQ4	Cat-	abun-	treat-	af-	(Intercept)	-	19359.257	-	0.999	NA	-	37923.044
	fish	dance	ment	ter		20.402	0.001				37963.849	
M4TABQ4	Cat-	biomas-	treat-	af-	(Intercept)	3.571	2.008	1.778	0.075	NA	-	7.508
	fish		ment	ter							0.365	
M4TABQ4	Cat-	biomas-	treat-	af-	Site_type_de-	-	4.577	-	0.160	NA	-	2.539
	fish		ment	ter	tailStone_pile	6.432	1.405				15.402	
M4TABQ4	Cat-	biomas-	treat-	af-	monitoring_int	-	0.688	-	0.209	NA	-	0.484
	fish		ment	ter		0.864	1.256				2.212	
M4TABQ4	Cat-	biomas-	treat-	af-	Site_type_de-	1.928	1.214	1.588	0.112	0.112	-	4.307
	fish		ment	ter	tail-						0.451	
					Stone_pile:mon-							
					itoring_int							
M4RABQ4	Kōura	abun-	ref-	af-	(Intercept)	-	0.976	-	0.448	NA	-	1.173
		dance	er-	ter		0.740	0.758				2.652	
			ence									
M4RABQ4	Kōura	abun-	ref-	af-	Site_type_de-	-	0.601	-	0.710	NA	-	0.955
		dance	er-	ter	tailRefer-	0.223	0.372				1.402	
			ence		ence_2							
M4RABQ4	Kōura	abun-	ref-	af-	monitoring_int	-	0.094	-	0.072	NA	-	0.015
		dance	er-	ter		0.169	1.801				0.353	
			ence									
M4RABQ4	Kōura	abun-	ref-	af-	Temperature	0.071	0.034	2.091	0.037	NA	0.004	0.138
		dance	er-	ter								
			ence									
M4RABQ4	Kōura	abun-	ref-	af-	Site_type_de-	-	0.124	-	0.761	0.902	-	0.206
		dance	er-	ter	tailRefer-	0.038	0.304				0.281	
			ence		ence_2:moni-							
					toring_int							

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Re- sponse	com- pari- type	time	frame	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- B	low	high	
M4RA04	Kōura	biomass	ref- er- ence	af- ter	(Intercept)	3.370	0.860	3.917	0.000	NA	1.683	5.056
M4RA04	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2	- 0.184	0.737 0.250	- 0.803	NA	- 1.630	-	1.261
M4RA04	Kōura	biomass	ref- er- ence	af- ter	monitoring_int	- 0.104	0.117 0.891	- 0.373	NA	- 0.334	-	0.125
M4RA04	Kōura	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	- 0.040	0.167 0.237	- 0.812	0.902	- 0.368	-	0.288
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	(Intercept)	- 6.715	1.788 3.757	- 0.000	NA	- 10.219	-	3.212
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2	0.731	1.033 0.707	0.479	NA	- 1.295	-	2.756
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	monitoring_int	- 0.051	0.233 0.220	- 0.826	NA	- 0.509	-	0.406
M4RA04	Cat- fish	abun- dance	ref- er- ence	af- ter	Temperature	0.258	0.082	3.131	0.002	NA	0.096	0.419

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Re- Q1	species	response	com- pari- type	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- B	low	high
M4RAQ4	Cat-	abun-	ref-	af-	Site_type_de-	-	0.296	-	0.623	0.902	-	0.435	
	fish	dance	er-	ter	tailRefer-	0.146		0.491			0.727		
			ence		ence_2:moni-								
					toring_int								
M4RAQ4	Cat-	abun-	ref-	af-	(Intercept)	-	13711.191	-	0.999	NA	-	26853.421	
	fish	dance	er-	ter		20.018		0.001			26893.458		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	(Intercept)	2.128	1.527	1.393	0.163	NA	-	5.121	
	fish		er-	ter							0.865		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	Site_type_de-	0.062	1.575	0.039	0.969	NA	-	3.149	
	fish		er-	ter	tailRefer-						3.025		
			ence		ence_2								
M4RAQ4	Cat-	biomass	ref-	af-	monitoring_int	0.007	0.325	0.023	0.982	NA	-	0.644	
	fish		er-	ter							0.629		
			ence										
M4RAQ4	Cat-	biomass	ref-	af-	Site_type_de-	0.055	0.448	0.123	0.902	0.902	-	0.933	
	fish		er-	ter	tailRefer-						0.823		
			ence		ence_2:moni-								
					toring_int								



Figure 1: Map of the monitoring sites.

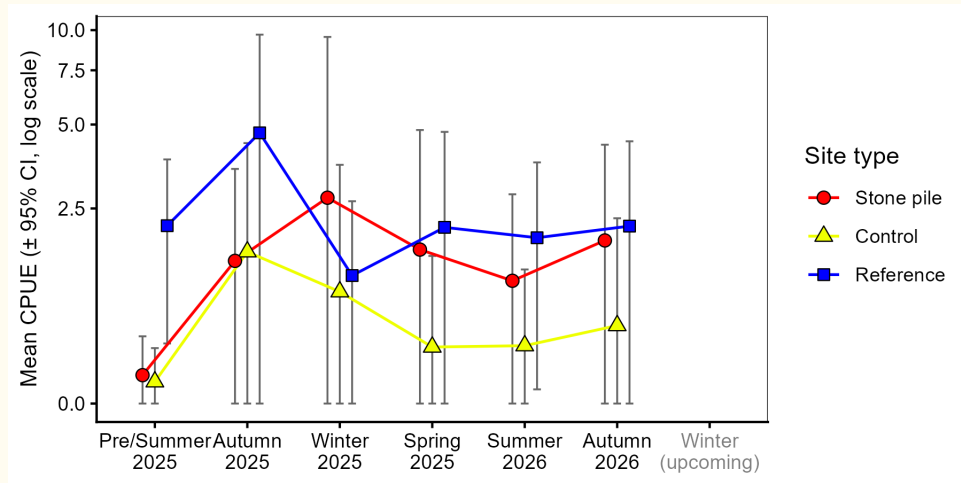


Figure 2: Mean catch per monitoring round ($\pm$ 95% CI) by site type for kōura.

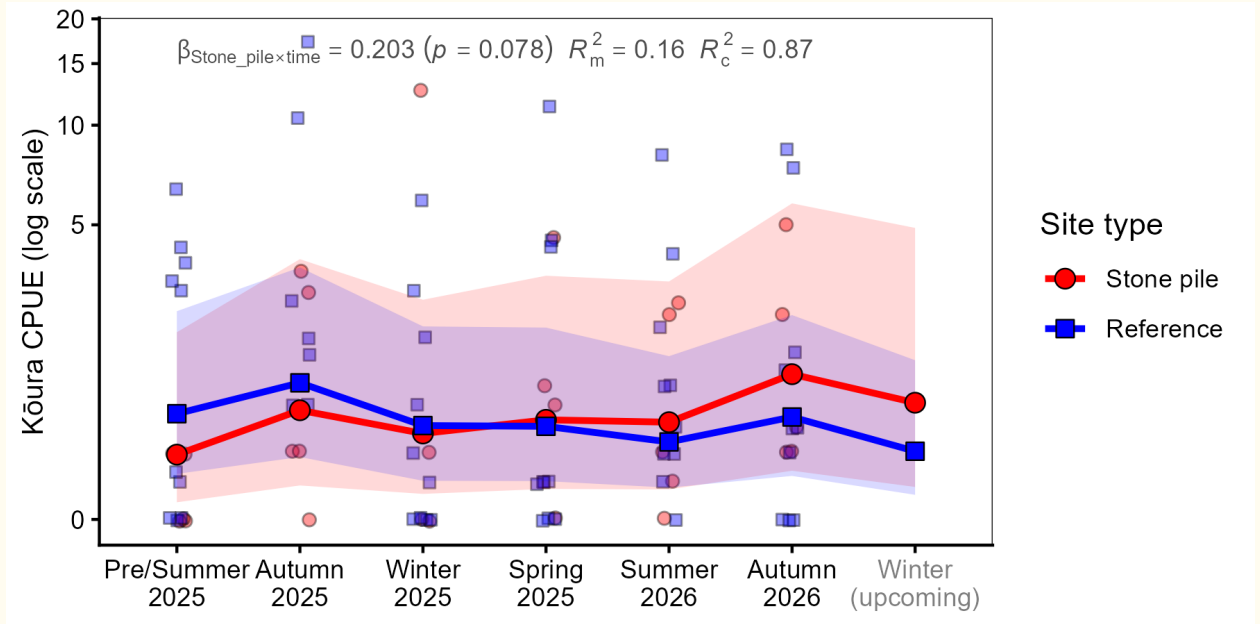


Figure 3: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the RQ3 abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\pm$ p-value) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

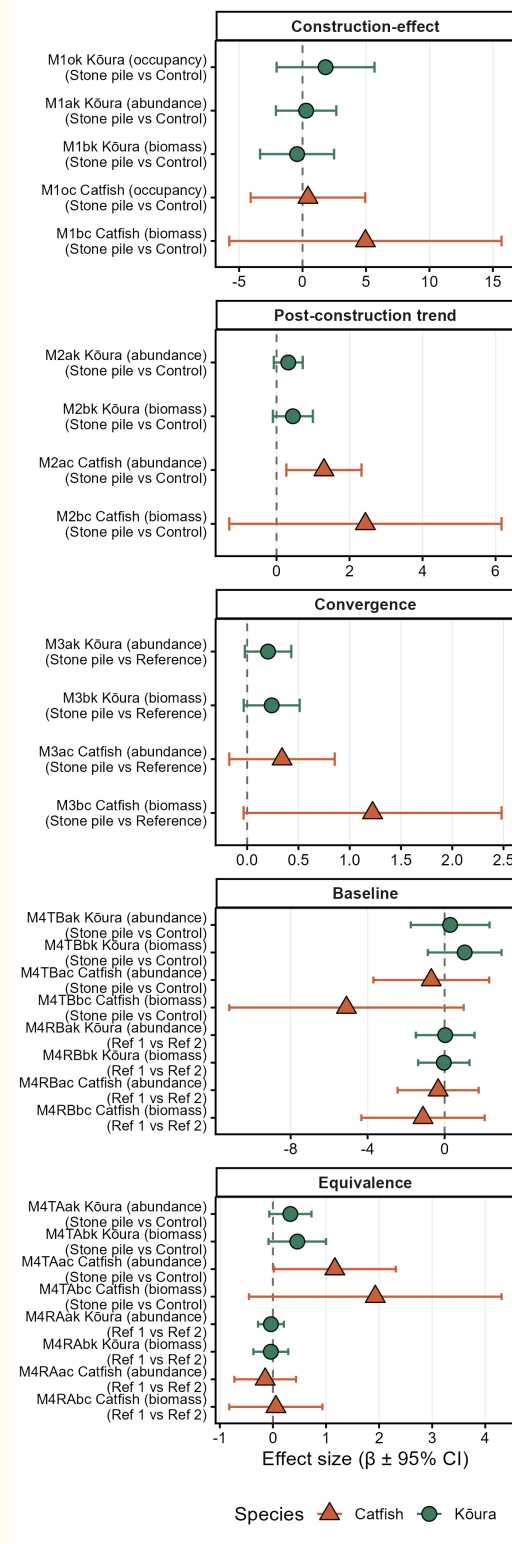


Figure 4: Effect sizes (β ± 95% CI) for the key term of interest of every model in model_registry (kōura and catfish, RQ1–RQ4). Filled points indicate significant effects (p < 0.05). Each panel has its own x-axis scale— magnitudes are not directly comparable across panels because each RQ tests a different contrast (step-change, per-round slope, or simple group difference) and RQ1 occupancy is on the log-odds scale while all others use the Tweedie log link. Models with degenerate confidence intervals (non-positive-definite Hessian) are excluded.

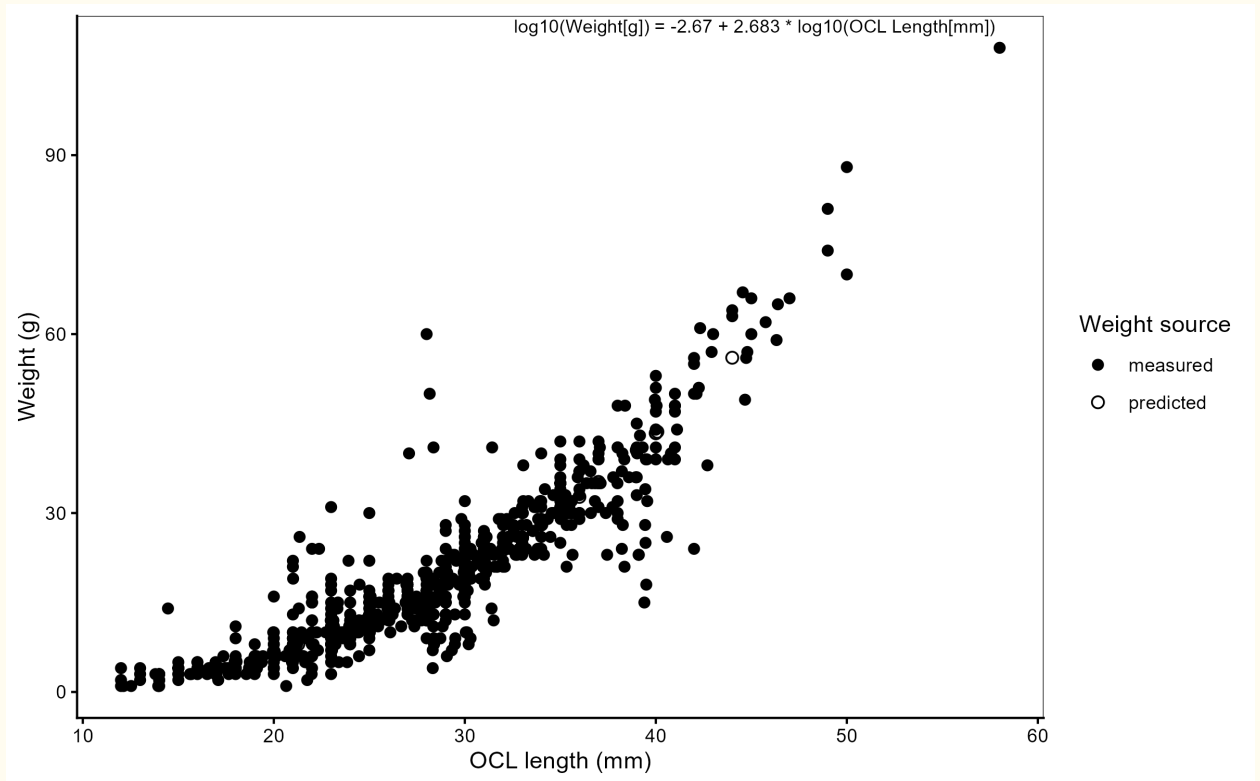


Figure 5: Length–weight relationship for kōura, fitted on log –log scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

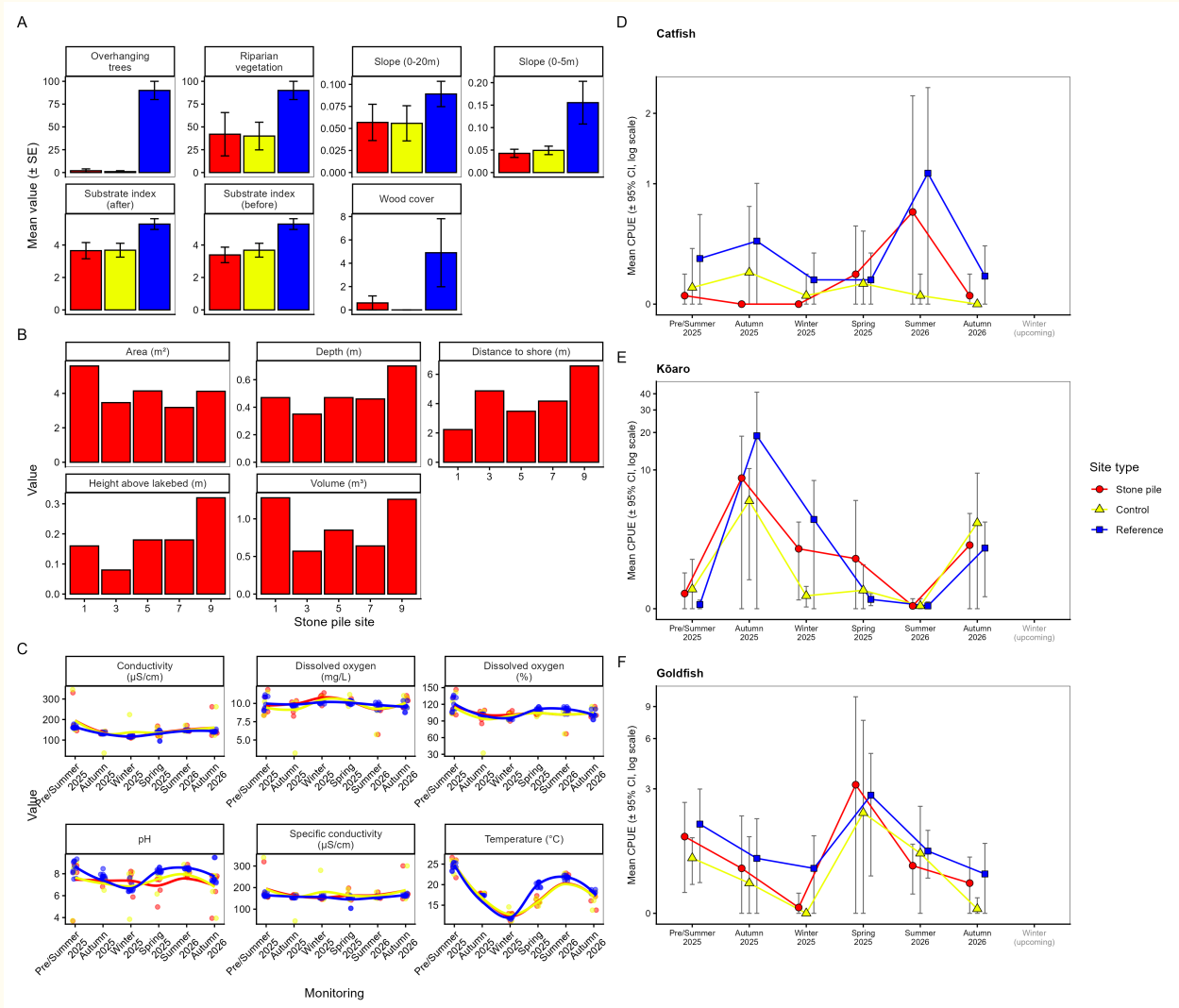


Figure 6: Descriptive context plots, combined. Left column (top to bottom): baseline habitat characteristics (mean $\pm$ SE by site type), stone pile structure dimensions (5 stone pile sites), chemical conditions over time (mean $\pm$ smoothed trend by site type). Right column: mean catch per monitoring round by site type, for catfish, kōaro, and goldfish. Site-type colours are shared across all panels — see the legend on the right. Background/methods-supporting figures, not primary results — see Results for the BACI, trend, and convergence models.

Table 3: DHARMA goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M1ok	0.115	0.972	1
M1ak	0.791	0.912	1
M1bk	0.453	0.810	1
M1oc	0.504	0.700	1
M1bc	0.546	0.406	1
M2ak	0.326	0.736	1
M2bk	0.633	0.720	1
M2ac	0.481	0.208	1
M2bc	0.400	0.186	1
M3ak	0.856	0.626	1
M3bk	0.972	0.602	1
M3ac	0.473	0.374	1
M3bc	0.580	0.260	1
M4TAak	0.416	0.970	1
M4TAbk	0.825	0.866	1
M4TAbc	0.697	0.668	1
M4RAak	0.956	0.874	1
M4RAbk	0.872	0.896	1
M4RAbc	0.885	0.748	1

²⁰ Source: [Article Notebook](#)

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